

■ Learning Check

Simple quizzes such as this one are included at the end of most sections. Though not designed to be a thorough examination of the prior material, if you have trouble answering these questions, it probably means you should go back and reread the section. Answers are given at the end of each Learning Check.

- Three of the fundamental physical quantities in physics are
 - distance, time, and weight
 - distance, time, and mass

- distance, time, and speed
 - distance, area, and volume
- (True or False.) When the period of a pendulum is less than 1 second, the frequency is always greater than 1 hertz.
 - One kilogram is the same as _____ grams.
 - (True or False.) In mechanics, weight and mass measure the same property of matter.

ANSWERS: 1. (b) 2. True 3. 1,000 4. False

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- For objects moving with constant speed,
 - the instantaneous speed is always zero.
 - the average speed is always zero.
 - the instantaneous speed and average speed are always equal.
 - None of the above.
- The highest speed that can be observed in Nature is
 - the speed of light.
 - the speed of sound.
 - 1,000,000 m/s.
 - There is no limit.
- A physical quantity that has both a magnitude and a direction associated with it is called a _____.

- (True or False.) As a car is traveling down a road, it is possible for its speed to be changing while its velocity is constant.
- (Choose the *incorrect* statement.) When two velocities are added, the resultant velocity
 - can be parallel to one or both of the original velocities.
 - cannot have zero magnitude.
 - can have a greater magnitude than either of the original velocities.
 - can be perpendicular to one of the original velocities.

ANSWERS: 1. (c) 2. (a) 3. vector 4. False 5. (b)

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- When an object's velocity is changing, we say that the object is _____.
- A freely falling body experiences
 - constant velocity.
 - centripetal acceleration.
 - constant acceleration.
 - steadily increasing acceleration.
- (True or False.) The acceleration of a slow-moving object can be greater than the acceleration of a fast-moving object.

- When something moves in a circle with constant speed,
 - its acceleration is perpendicular to its velocity.
 - its acceleration is zero.
 - its velocity is constant.
 - its acceleration is parallel to its velocity.
- An acceleration of 2.5 *g* is equivalent to _____ m/s².

ANSWERS: 1. accelerating 2. (c) 3. True 4. (a) 5. 24.5

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1. What is different about the graph of distance versus time for a fast-moving object compared to that of a slow-moving object?
2. (True or False.) A rock dropped from the top of a tall building travels the same distance during each 1-second interval that it falls.
3. If the graph of a car's velocity versus time curves upward, we can conclude its acceleration is _____.
4. Graphs of distance versus time and velocity versus time are useful for analyzing
 - (a) the motion of a falling body.
 - (b) the performance of a car.
 - (c) laboratory measurements of an accelerating object.
 - (d) All of the above.

ANSWERS: 1. larger slope 2. false 3. increasing 4. (d)